Algorithms for lattice fermions (ALF): a flexible package for the efficient simulation of lattice Hamiltonians

Julius-Maximilians-
UNIVERSITÄT WÜRZBURG

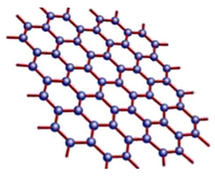
The ALF collaboration (F. F. Assaad, M. Bercx, F. Goth, J. S. Hofmann and J. Schwab)

SFB1170 ToCoTronics

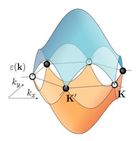
Goals:

- ✓ Generic package for Monte Carlo simulations of fermion-boson lattice models
- ✓ Efficient implementation on modern HPC systems
- ✓ Benchmarking and golden standards

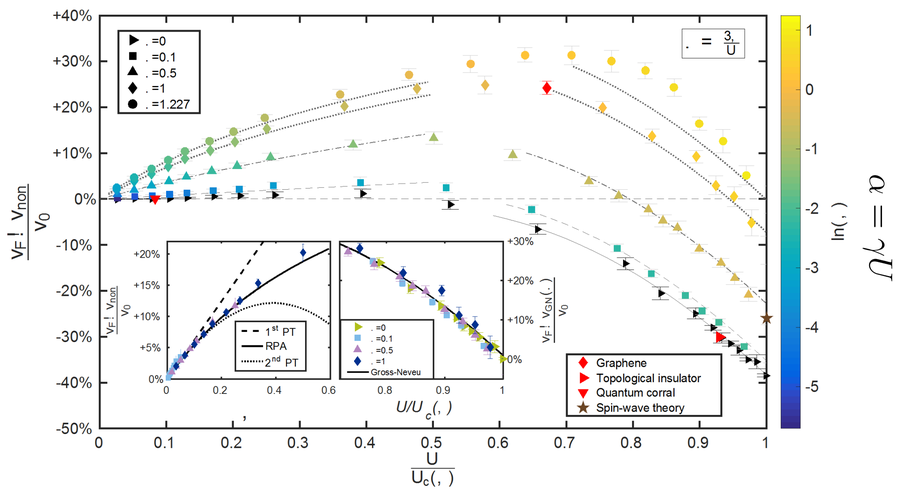
The role of electron-electron interactions in two dimensional Dirac fermions



$$\hat{H} = -t \sum_{\langle i,j\rangle,\sigma=1}^{2} \left(\hat{c}^{\dagger}_{i,\sigma}\hat{c}_{j,\sigma} + \hat{c}^{\dagger}_{j,\sigma}\hat{c}_{i,\sigma}\right) + \sum_{i,j} (\hat{n}_i - 1)\, V_{i,j}\, (\hat{n}_j - 1)$$



$$V_{i,j} = U \begin{cases} 1 & i = j \\ \frac{\gamma}{r_{i,j}} & i \neq j \end{cases}$$



Ho-Kin Tang, J. N. Leaw, J. N. B. Rodrigues, I. F. Herbut, P. Sengupta, F. F. Assaad, and S. Adam, Science, to appear

PHYSICAL REVIEW LETTERS **120,** 107201 (2018)

Quantum Monte Carlo Simulation of Frustrated Kondo Lattice Models

Toshihiro Sato,[1] Fakher F. Assaad,[1] and Tarun Grover[2]

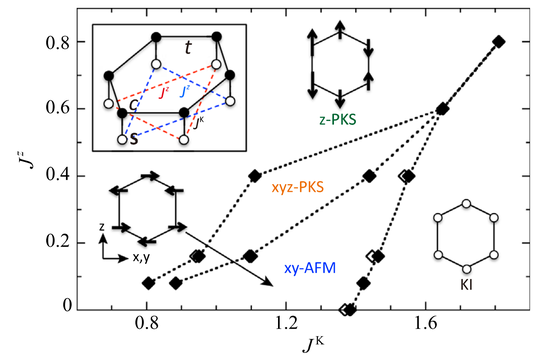
$$\hat{H} = \hat{H}_{\mathrm{Spin}} + \hat{H}_{\mathrm{Fermion}} + \hat{H}_{\mathrm{Kondo}}$$

$$\hat{H}_{\mathrm{Spin}} = \sum_{i,j}[J^{z}_{ij}\hat{S}^{z}_{i}\hat{S}^{z}_{j} + J^{\perp}_{ij}(\hat{S}^{+}_{i}\hat{S}^{-}_{j} + \mathrm{H.c.})],$$

$$\hat{H}_{\mathrm{Fermion}} = \sum_{x,y,\sigma}\hat{c}^{\dagger}_{x\sigma}T_{x,y}\hat{c}_{y\sigma} + \sum_{x}U\left(\hat{n}_{x,\downarrow} - \frac{1}{2}\right)\left(\hat{n}_{x,\uparrow} - \frac{1}{2}\right),$$

$$\hat{H}_{\mathrm{Kondo}} = \sum_{i,x}\frac{J^{K}_{i,x}}{2}\hat{\boldsymbol{c}}^{\dagger}_{x}[\sigma^{z}\cdot\hat{S}^{z}_{i} - (-1)^{x}(\sigma^{+}\hat{S}^{-}_{i} + \sigma^{-}\hat{S}^{+}_{i})]\hat{\boldsymbol{c}}_{x}.$$

Interplay of Kondo, RKKY and geometric frustration



The ALF package

$$\hat{H} = \sum_{k=1}^{M_T}\sum_{\sigma=1}^{N_{\mathrm{col}}}\sum_{s=1}^{N_{\mathrm{fl}}}\sum_{x,y}^{N_{\mathrm{dim}}}\hat{c}^{\dagger}_{x\sigma s}T^{(ks)}_{xy}\hat{c}_{y\sigma s} + \sum_{k=1}^{M_V}U_k\left\{\sum_{\sigma=1}^{N_{\mathrm{col}}}\sum_{s=1}^{N_{\mathrm{fl}}}\left[\left(\sum_{x,y}^{N_{\mathrm{dim}}}\hat{c}^{\dagger}_{x\sigma s}V^{(ks)}_{xy}\hat{c}_{y\sigma s}\right) + \alpha_{ks}\right]\right\}^{2} + \sum_{k=1}^{M_I}\hat{Z}_k\left(\sum_{\sigma=1}^{N_{\mathrm{col}}}\sum_{s=1}^{N_{\mathrm{fl}}}\sum_{x,y}^{N_{\mathrm{dim}}}\hat{c}^{\dagger}_{x\sigma s}I^{(ks)}_{xy}\hat{c}_{y\sigma s}\right) + \hat{H}_{\mathrm{Ising}}$$

Kinetic — Potential (sum of perfect squares) — Coupling of fermions to Ising field

- ➢ Model can be specified at minimal programming cost
- ➢ MPI/OpenMP implementation
- ➢ Fortran 2003 standard
- ➢ SU(N_{col}) symmetric in colors N_{col}
- ➢ Arbitrary Bravais lattice for d=1, 2
- ➢ Block diagonal in flavors, N_{fl}
- ➢ Parallel tempering, projective and finite T approaches

M. Bercx, F. Goth, J. S. Hofmann and F. F. Assaad, SciPost Phys. 3 (2017), 013.

Future developments

- ➢ Continuous fields
- ➢ Hybrid molecular dynamics
- ➢ Langevin dynamics

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DFG Wissenschaftliche Literaturversorgungs und Informationssysteme (LIS)

PNAS

Confinement transition of $\mathbb{Z}_2$ gauge theories coupled to massless fermions: Emergent quantum chromodynamics and *SO*(5) symmetry

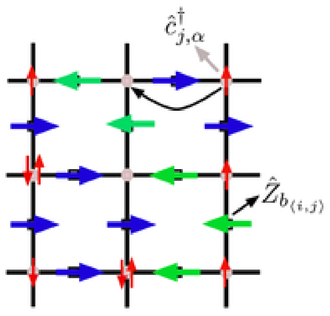
Snir Gazit[a,1], Fakher F. Assaad[b], Subir Sachdev[c,d,1], Ashvin Vishwanath[c], and Chong Wang[c]

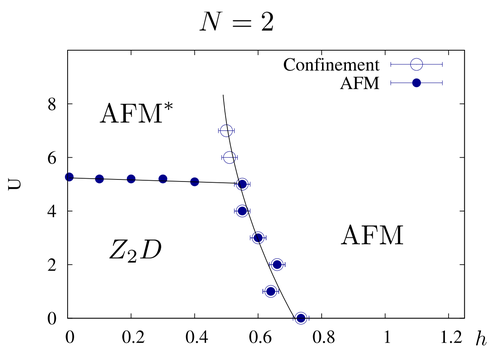
PHYSICAL REVIEW X **6,** 041049 (2016)

Simple Fermionic Model of Deconfined Phases and Phase Transitions

F. F. Assaad[1] and Tarun Grover[2,3]

$$\hat{H} = \sum_{\langle i,j\rangle}\hat{Z}_{\langle i,j\rangle}\left(\sum_{\alpha=1}^{N}\hat{c}^{\dagger}_{i,\alpha}\hat{c}_{j,\alpha} + \mathrm{H.c.}\right) - Nh\sum_{\langle i,j\rangle}\hat{X}_{\langle i,j\rangle}.$$





	Gauge Field	**Particle-hole**
Z_2D	De-confined	No binding
AFM	Confined	Condensed
AFM*	De-confined	Condensed

The c-electrons carry global U(1) charge and local Z_2 charge

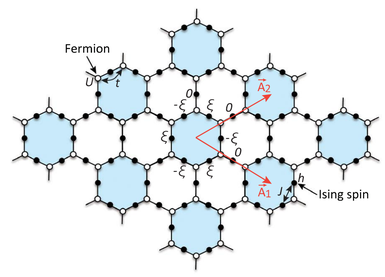
PRL **119,** 197203 (2017) PHYSICAL REVIEW LETTERS week ending 10 NOVEMBER 2017

Dirac Fermions with Competing Orders: Non-Landau Transition with Emergent Symmetry

Toshihiro Sato,[1] Martin Hohenadler,[1] and Fakher F. Assaad[1]

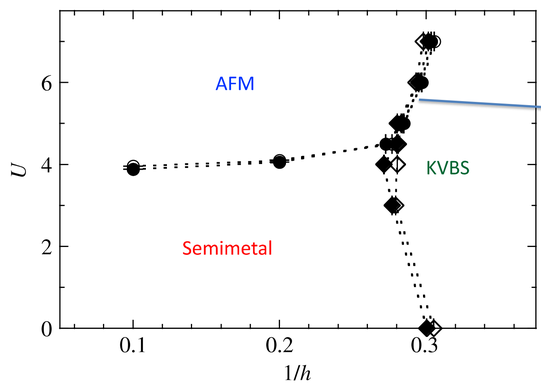
$$\hat{H} = \hat{H}_f + \hat{H}_s + \hat{H}_{fs}.$$

$$\hat{H}_f = -t\sum_{\langle ij\rangle,\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + U\sum_{i}\left(\hat{n}_{i\uparrow} - \frac{1}{2}\right)\left(\hat{n}_{i\downarrow} - \frac{1}{2}\right)$$



$$\hat{H}_s = J\sum_{\langle ij,kl\rangle}\hat{s}^{z}_{ij}\hat{s}^{z}_{kl} - h\sum_{\langle ij\rangle}\hat{s}^{x}_{ij}$$

$$\hat{H}_{fs} = \sum_{\langle ij\rangle,\sigma}\xi_{ij}\hat{s}^{z}_{ij}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}.$$



Direct and continuous transition with emergent O(4) symmetry

Project Z03